Comparative Analysis of Real-Time Object Detection Algorithms on Colonoscopy Video Datasets

Master Thesis

Master of Science in Applied Computer Science

Andreas Franz Schwab

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Supervisor:

1st: ProfḊrĊhristian Ledig
2nd: M.Sc Sebastian Dörrich

Chair of Explainable Machine Learning
Faculty of Information Systems and Applied Computer Sciences
Otto-Friedrich-University Bamberg

Abstract

Short summary of your thesis (max. 1 page) ...

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List of Acronyms

AI Artificial Intelligence

Notation

Please list all mathematical notations that are used in the thesis here. For reference, we provide you with an example from the book by Goodfellow et al. (2016).

Numbers and Arrays

a	A scalar (integer or real)
$\boldsymbol{a}$	A vector
$\boldsymbol{A}$	A matrix
$\mathsf{\mathbf{A}}$	A tensor
$\boldsymbol{I}_n$	Identity matrix with n rows and n columns
$\boldsymbol{I}$	Identity matrix with dimensionality implied by context
$\boldsymbol{e}^{(i)}$	Standard basis vector $[0, \ldots, 0, 1, 0, \ldots, 0]$ with a 1 at position i
$\text{diag}(\boldsymbol{a})$	A square, diagonal matrix with diagonal entries given by $\boldsymbol{a}$
a	A scalar random variable
$\mathbf{a}$	A vector-valued random variable
$\mathbf{A}$	A matrix-valued random variable

Sets and Graphs

$\mathbb{A}$	A set
$\mathbb{R}$	The set of real numbers
$\{0, 1\}$	The set containing 0 and 1
$\{0, 1, \ldots, n\}$	The set of all integers between 0 and n
$[a, b]$	The real interval including a and b
$(a, b]$	The real interval excluding a but including b
$\mathbb{A} \backslash \mathbb{B}$	Set subtraction, i.e., the set containing the elements of $\mathbb{A}$ that are not in $\mathbb{B}$
$\mathcal{G}$	A graph
$Pa_{\mathcal{G}}(\mathrm{x}_i)$	The parents of x_i in $\mathcal{G}$

Indexing

a_i	Element i of vector $\boldsymbol{a}$, with indexing starting at 1
a_{-i}	All elements of vector $\boldsymbol{a}$ except for element i
$A_{i,j}$	Element i, j of matrix $\boldsymbol{A}$
$\boldsymbol{A}_{i,:}$	Row i of matrix $\boldsymbol{A}$
$\boldsymbol{A}_{:,i}$	Column i of matrix $\boldsymbol{A}$
$\mathsf{A}_{i,j,k}$	Element (i, j, k) of a 3-D tensor $\mathbf{\mathsf{A}}$
$\mathbf{\mathsf{A}}_{:,:,i}$	2-D slice of a 3-D tensor
a_i	Element i of the random vector $\mathbf{a}$

Linear Algebra Operations

$\boldsymbol{A}^\top$	Transpose of matrix $\boldsymbol{A}$
$\boldsymbol{A}^+$	Moore-Penrose pseudoinverse of $\boldsymbol{A}$
$\boldsymbol{A} \odot \boldsymbol{B}$	Element-wise (Hadamard) product of $\boldsymbol{A}$ and $\boldsymbol{B}$
$\det(\boldsymbol{A})$	Determinant of $\boldsymbol{A}$

Calculus

$\frac{dy}{dx}$	Derivative of y with respect to x
$\frac{\partial y}{\partial x}$	Partial derivative of y with respect to x
$\nabla_{\boldsymbol{x}} y$	Gradient of y with respect to $\boldsymbol{x}$
$\nabla_{\boldsymbol{X}} y$	Matrix derivatives of y with respect to $\boldsymbol{X}$
$\nabla_{\mathbf{\mathsf{X}}} y$	Tensor containing derivatives of y with respect to $\mathbf{\mathsf{X}}$
$\frac{\partial f}{\partial \boldsymbol{x}}$	Jacobian matrix $\boldsymbol{J} \in \mathbb{R}^{m \times n}$ of $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$
$\nabla_{\boldsymbol{x}}^2 f(\boldsymbol{x})$ or $\boldsymbol{H}(f)(\boldsymbol{x})$	The Hessian matrix of f at input point $\boldsymbol{x}$
$\int f(\boldsymbol{x}) d\boldsymbol{x}$	Definite integral over the entire domain of $\boldsymbol{x}$
$\int_{\mathbb{S}} f(\boldsymbol{x}) d\boldsymbol{x}$	Definite integral with respect to $\boldsymbol{x}$ over the set $\mathbb{S}$

Probability and Information Theory

$\mathrm{a} \bot \mathrm{b}$	The random variables a and b are independent
$\mathrm{a} \bot \mathrm{b} \mid \mathrm{c}$	They are conditionally independent given c
$P(\mathrm{a})$	A probability distribution over a discrete variable
$p(\mathrm{a})$	A probability distribution over a continuous variable, or over a variable whose type has not been specified
$\mathrm{a} \sim P$	Random variable a has distribution P
$\mathbb{E}_{\mathrm{x} \sim P}[f(x)]$ or $\mathbb{E} f(x)$	Expectation of $f(x)$ with respect to $P(\mathrm{x})$
$\mathrm{Var}(f(x))$	Variance of $f(x)$ under $P(\mathrm{x})$
$\mathrm{Cov}(f(x), g(x))$	Covariance of $f(x)$ and $g(x)$ under $P(\mathrm{x})$
$H(\mathrm{x})$	Shannon entropy of the random variable x
$D_{\mathrm{KL}}(P \Vert Q)$	Kullback-Leibler divergence of P and Q
$\mathcal{N}(\boldsymbol{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$	Gaussian distribution over $\boldsymbol{x}$ with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$

Functions

$f : \mathbb{A} \rightarrow \mathbb{B}$	The function f with domain $\mathbb{A}$ and range $\mathbb{B}$
$f \circ g$	Composition of the functions f and g
$f(\boldsymbol{x}; \boldsymbol{\theta})$	A function of $\boldsymbol{x}$ parametrized by $\boldsymbol{\theta}$. (Sometimes we write $f(\boldsymbol{x})$ and omit the argument $\boldsymbol{\theta}$ to lighten notation)
$\log x$	Natural logarithm of x
$\sigma(x)$	Logistic sigmoid, $\dfrac{1}{1 + \exp(-x)}$
$\zeta(x)$	Softplus, $\log(1 + \exp(x))$
$\vert\vert \boldsymbol{x} \vert\vert_p$	L^p norm of $\boldsymbol{x}$
$\vert\vert \boldsymbol{x} \vert\vert$	L^2 norm of $\boldsymbol{x}$
x^+	Positive part of x, i.e., $\max(0, x)$
$\mathbf{1}_{\mathrm{condition}}$	is 1 if the condition is true, 0 otherwise

Sometimes we use a function f whose argument is a scalar but apply it to a vector, matrix, or tensor: $f(\boldsymbol{x})$, $f(\boldsymbol{X})$, or $f(\mathsf{X})$. This denotes the application of f to the array element-wise. For example, if $\mathsf{C} = \sigma(\mathsf{X})$, then $\mathsf{C}_{i,j,k} = \sigma(\mathsf{X}_{i,j,k})$ for all valid values of i, j and k.

Datasets and Distributions

p_{data}	The data generating distribution
$\hat{p}_{\text{data}}$	The empirical distribution defined by the training set
$\mathbb{X}$	A set of training examples
$\boldsymbol{x}^{(i)}$	The i-th example (input) from a dataset
$y^{(i)}$ or $\boldsymbol{y}^{(i)}$	The target associated with $\boldsymbol{x}^{(i)}$ for supervised learning
$\boldsymbol{X}$	The $m \times n$ matrix with input example $\boldsymbol{x}^{(i)}$ in row $\boldsymbol{X}_{i,:}$

Etc

Some more of your text. For citations, use the command `\citep{lecun2015deep}` which produces (LeCun et al., 2015) or `\cite{lecun2015deep}` which produces LeCun et al. (2015). Table 1. Figure 1 shows ...

Table 1: Sample table title

Name	Description	Size (μm)
Dendrite	Input terminal	$\sim$100
Axon	Output terminal	$\sim$10
Soma	Cell body	up to 10^6

Figure 1: Logo of the chair of Explainable Machine Learning

If you want to typeset formulas, there is the inline version $y = x_1^{0.5} x_2^{0.5}$, centered like this

$$y = x_1^{0.5} x_2^{0.5}$$

or numbered:

$$y = x_1^{0.5} x_2^{0.5} \tag{1}$$

so that you can refer to equation 1 in the text.

1 Introduction

1. Introduction 1.1 Motivation Rising importance of early colorectal cancer detection via polyps. Real-time computer-assisted detection (CADe) as a clinical game-changer in colonoscopy.

1.2 Problem Statement Challenges in real-time detection: limited full-procedure datasets, small object sensitivity, high false positives.

1.3 Objectives and Scope Evaluate multiple state-of-the-art real-time object detectors on annotated colonoscopy videos. Examine trade-offs between detection accuracy, speed, and clinical applicability.

1.4 Research Questions Which real-time object detection algorithms best balance accuracy and inference speed for colonoscopy videos? How well do these models detect small or early-stage polyps? What impact do negative frames and class imbalance have on performance? Can 2D detections be leveraged for 3D spatial understanding in real-time?

1.5 Contributions Benchmark of detectors on full-length video datasets. Real-time inference application with visualization. In-depth analysis of detection failures, including small and missed polyps. Discussion of feasibility for 3D spatial reasoning using monocular endoscopy videos.

1.6 Structure of the Thesis Brief chapter overview.

2 Background and Related Work

2. Background and Related Work 2.1 Clinical Context Overview of colorectal cancer and clinical relevance of polyp detection. Endoscopy workflow and challenges in real-time interpretation.

2.2 CAD Systems in Endoscopy Difference between CADe (detection) and CADx (diagnosis). Regulatory considerations and AI integration into practice.

2.3 Object Detection in Computer Vision Detection pipeline fundamentals: backbone, neck, head. Categories of object detectors: One-stage vs. two-stage Anchor-based (YOLO, SSD) vs. anchor-free (FCOS, RT-DETR) Transformer-based models (DETR variants) Real-time constraints: trade-offs in model size, latency, and accuracy.

2.4 Related Work in Medical Object Detection Summary of benchmark studies in endoscopic polyp detection. Limitations of current evaluations (e.g., static image datasets, lack of video context).

3 Datasets

3. Datasets 3.1 Primary Dataset: REAL-Colon Description of dataset composition and diversity. Annotation format (COCO-style), bounding boxes, and categories. Negative frame ratio, dataset statistics, and baseline performance.

3.2 Supplemental Datasets Overview of public datasets used for validation: Kvasir, CVC-ClinicVideoDB, ASU-Mayo, SUN, LDPolypVideo. Differences in resolution, annotation styles, and video quality.

3.3 Preprocessing and Frame Sampling Frame extraction, balancing strategies (negative/positive), and annotation normalization. Data augmentation techniques used.

4 Methodology

4. Methodology 4.1 Selection of Detection Models YOLOv8, EfficientDet-D0/D1, FCOS, RT-DETR, SSD, NanoDet. Selection criteria: popularity, real-time capability, architecture diversity. 4.2 Implementation Details Frameworks (e.g., PyTorch, TensorFlow, MMDetection). Hardware setups and resource availability. Training and fine-tuning pipelines (image size, batch size, learning rate).

4.3 Evaluation Metrics Detection Metrics: mAP@[.5:.95], Precision, Recall, F1, IoU, Localization error. Performance Metrics: Inference time, FPS, model size, GPU usage, memory, FLOPs. Clinical Metrics: False Positive Rate (FPR), Detection Rate (DR), Missed small polyps.

4.4 Validation Strategy Dataset splits (train/val/test). Cross-validation or hold-out testing. Use of negative frames and impact on false positives.

5 Results and Analysis

5. Results and Analysis 5.1 Quantitative Results Tables comparing detection metrics across models. FPS vs. accuracy plots.

5.2 Qualitative Results Visual examples of successful detections, false positives, and missed cases. Small polyp detection analysis.

5.3 Comparative Discussion Trade-offs between real-time viability and precision. Suitability for clinical integration.

5.4 Ablation Studies (optional) Impact of batch size, input resolution, and augmentation on performance. Performance variance on small vs. large polyp detection.

6 Real-Time Visualization Tool

6. Real-Time Visualization Prototype 6.1 System Overview Implementation of real-time test app with overlay UI. Components: live feed, bounding box rendering, FPS counter.

6.2 Features and Design Interactive mode: pause, analyze frame, zoom. Export annotations or flagged detections.

6.3 Experimental: 3D Depth Estimation Simulated depth cues using motion or shape-based heuristics. Tools: Open3D, Blender, OpenCV. Alternative methods: Relative saliency maps, Depth ranking.

6.4 Hardware Performance Benchmarking Inference performance on various GPUs (V100, RTX A6000, 1070, RTX5000). Latency trade-offs across devices.

7 Discussion

7. Discussion 7.1 Summary of Key Findings Best performing models per use-case (real-time, accuracy, small polyp detection).

7.2 Clinical Impact and Considerations Potential of CADe in reducing missed polyps. Risk of false positives and over-reliance on AI.

7.3 Limitations Dataset variability, generalizability, annotation bias. Real-world latency vs. offline evaluation.

7.4 Reproducibility and Transparency Code availability, model checkpoints, experiment logs.

8 Conclusion

8. Conclusion and Future Work 8.1 Summary Recap of goals, methodology, and key insights. Research question answers.

8.2 Practical Implications Deployment potential, model recommendations for hospitals.

8.3 Future Directions Real-time instance segmentation and tracking. Clinical trials and user interface studies. Integration of depth estimation, stereo or NeRF pipelines. Multi-modal approaches combining endoscopy + CT or sensor data.

A Appendix

If needed for supplementary material, such as detailed description of data collection, tables, or figures.

Bibliography

Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep learning*. MIT press, 2016.

Yann LeCun, Yoshua Bengio, and Geoffrey Hinton. Deep learning. *nature*, 521 (7553):436–444, 2015.

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